102 Internal Energy, Heat and Heat Transfer

1.2.1 Temperature in the particle model

Temperature is related to the average kinetic energy of random particle motion. A higher temperature means more vigorous motion on average, not that every particle has the same speed. For an ideal gas, mean translational kinetic energy is proportional to temperature in kelvin; Celsius values cannot be used directly in this proportionality.

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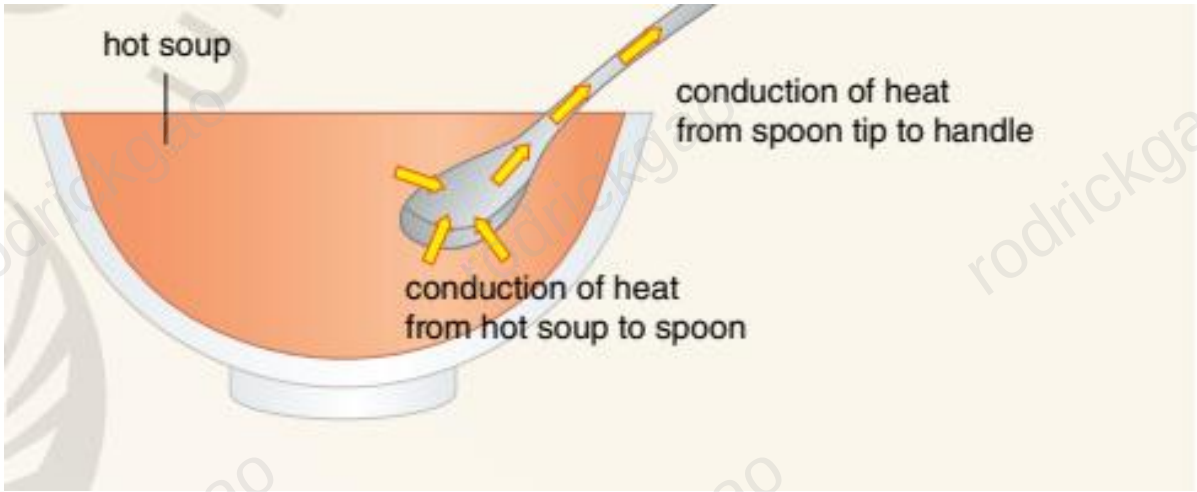
1.2.2 Heat and internal energy

Heat is energy transferred because of a temperature difference, naturally flowing from hot to cold. Internal energy is the total microscopic energy of the particles within an object. Absorbing or releasing heat generally changes internal energy, but heat describes a transfer rather than an amount stored as a property of the object.

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1.2.3 Conduction

Conduction transfers energy from hotter to colder regions without bulk movement of the material. Particle collisions and interactions transfer energy, while free electrons also play an important role in metals. Heat can therefore pass between objects in direct contact; microscopic vibration is not the same as the whole material flowing.

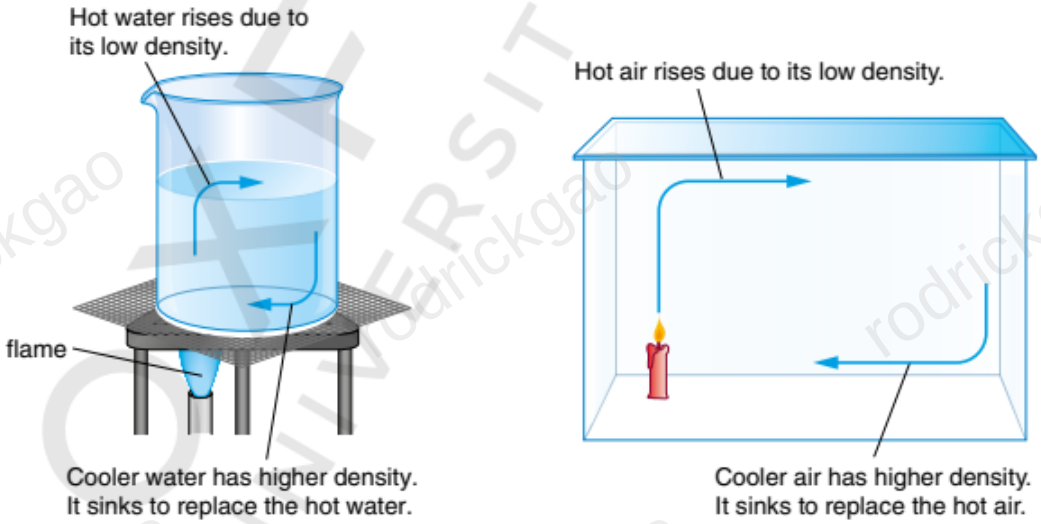


Heat travels along the spoon from the hot end to the cold end without bulk motion of the spoon.

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1.2.4 Convection

Convection transfers energy through bulk movement of a liquid or gas. In natural convection, heated fluid expands, becomes less dense and rises, while cooler, denser fluid sinks to replace it. Fans and pumps produce forced convection. Solids cannot transfer heat by this kind of bulk flow.



Convection in liquids and gases: warmer fluid rises and cooler fluid moves in to replace it.

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1.2.5 Thermal radiation

Objects emit and absorb energy as electromagnetic radiation. This transfer needs no medium and can cross a vacuum; at ordinary temperatures, infrared radiation is especially important. An object colder than its surroundings still emits radiation, but it may absorb more power than it emits and therefore gain energy overall.

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1.2.6 Surface properties and radiation

Under comparable conditions, dull black surfaces are generally good absorbers and emitters of thermal radiation. Light-coloured, shiny surfaces usually absorb and emit less and reflect more. Keep surface area and temperature controlled when comparing surfaces, so that colour and finish are not confused with area or temperature effects.

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1.2.7 Temperature difference and net radiation

For an object with fixed surface properties and area in a fixed environment, increasing its temperature difference from the surroundings generally increases the magnitude of net radiative heat transfer. Distinguish emitted power, absorbed power and their difference. This does not mean that the material's emissivity is directly proportional to the temperature difference.

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1.2.8 Surface area and radiative power

With the same surface properties, temperature and illumination, a larger area emits or absorbs more total radiative power because more surface exchanges energy. Distinguish total power from power per unit area: increasing area does not, by itself, increase the transfer per square metre.

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